NAME	Sheheryar Shakeel
ROLL NO	26K-3098

1. A university has 2 sections, each containing 3 students. Each student has marks in 4 subjects. Store the marks in a 3D array and write a C program to display the total and average marks of each student, find the highest-scoring student in each section, count students whose average is at least 50, and identify the section with the highest overall average.

3d.c | 3D_2.c | hospital.c | f1.c | f2.c | f3.c | f4.c | f5.c | f6.c | f7.c

folderofc.c > practice.c > 3d.c

```c
#include <stdio.h>
int main(){
    int marks[2][3][4];
    int total, count = 0;
    float avg, sectionTotal[2] = {0};
    float sectionAvg[2];
    int highest, highestStudent;
    for(int i = 0; i < 2; i++){
        printf("\nSection %d\n", i + 1);
        for(int j = 0; j < 3; j++){
            printf("\nStudent %d\n", j + 1);
            for(int k = 0; k < 4; k++){
                printf("Enter marks of subject %d: ", k + 1);
                scanf("%d", &marks[i][j][k]);
            }
        }
    }
    for(int i = 0; i < 2; i++){
        highest = -1;
        highestStudent = 0;
        sectionTotal[i] = 0;
        printf("\nSection %d Results\n", i + 1);
        for(int j = 0; j < 3; j++){
            total = 0;
            for(int k = 0; k < 4; k++){
                total = total + marks[i][j][k];
            }
            avg = total / 4.0;
            sectionTotal[i] = sectionTotal[i] + total;
            printf("Student %d: Total = %d, Average = %.2f\n",j + 1, total, avg);
            if(avg >= 50){
                count++;
            }
            if(total > highest){
                highest = total;
                highestStudent = j + 1;
            }
        }
        sectionAvg[i] = sectionTotal[i] / 12.0;
        printf("Highest-scoring student: Student %d\n",
               highestStudent);
        printf("Section average: %.2f\n", sectionAvg[i]);
    }
    printf("\nStudents with average at least 50: %d\n", count);
    if(sectionAvg[0] > sectionAvg[1]){
        printf("Section 1 has the highest overall average.");
    }
    else if(sectionAvg[1] > sectionAvg[0]){
        printf("Section 2 has the highest overall average.");
    }
    else{
        printf("Both sections have the same overall average.");
    }
    return 0;
}
```

```
Section 1

Student 1
Enter marks of subject 1: 12
Enter marks of subject 2: 14
Enter marks of subject 3: 15
Enter marks of subject 4: 13

Student 2
Enter marks of subject 1: 12
Enter marks of subject 2: 18
Enter marks of subject 3: 14
Enter marks of subject 4: 16

Student 3
Enter marks of subject 1: 12
Enter marks of subject 2: 15
Enter marks of subject 3: 14
Enter marks of subject 4: 12

Section 2

Student 1
Enter marks of subject 1: 13
Enter marks of subject 2: 21
Enter marks of subject 3: 25
Enter marks of subject 4: 24

Student 2
Enter marks of subject 1: 23
Enter marks of subject 2: 15
Enter marks of subject 3: 29
Enter marks of subject 4: 14

Student 3
Enter marks of subject 1: 17
Enter marks of subject 2: 15
Enter marks of subject 3: 17
Enter marks of subject 4: 18

Section 1 Results
Student 1: Total = 54, Average = 13.50
Student 2: Total = 60, Average = 15.00
Student 3: Total = 53, Average = 13.25
Highest-scoring student: Student 2
Section average: 13.92

Section 2 Results
Student 1: Total = 83, Average = 20.75
Student 2: Total = 81, Average = 20.25
Student 3: Total = 67, Average = 16.75
Highest-scoring student: Student 1
Section average: 19.25

Students with average at least 50: 0
Section 2 has the highest overall average.
PS C:\Users\I C\VS\folderofc.c\practice.c>
```

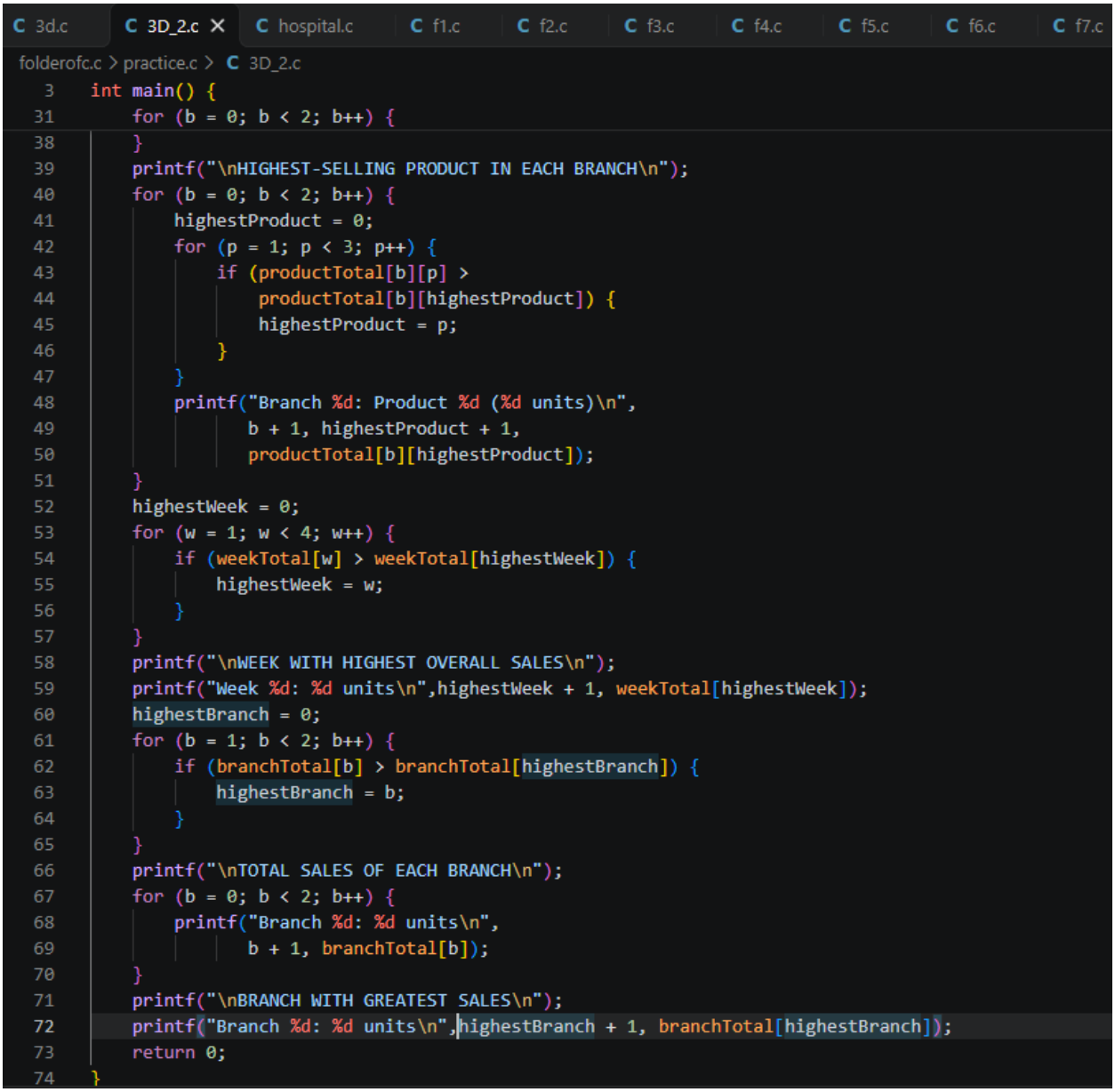
2. A company has 2 branches, sells 3 products, and records sales quantities for 4 weeks. Store the sales data in a 3D array and write a C program to calculate the total quantity sold for each product in each branch, identify the product with the highest sales in each branch, find the week with the highest overall sales, and determine which branch sold the greatest quantity of products.

folderofc.c > practice.c > C 3D_2.c

```c

#include <stdio.h>
int main() {
    int sales[2][3][4] = {
        {
            {10, 15, 20, 25},
            {12, 18, 22, 28},
            {8, 14, 16, 20}
        },
        {
            {15, 20, 25, 30},
            {18, 22, 28, 32},
            {10, 16, 20, 24}
        }
    };
    int branchTotal[2] = {0, 0};
    int weekTotal[4] = {0, 0, 0, 0};
    int productTotal[2][3] = {0};
    int b, p, w;
    int highestProduct, highestWeek, highestBranch;
    for (b = 0; b < 2; b++) {
        for (p = 0; p < 3; p++) {
            for (w = 0; w < 4; w++) {
                productTotal[b][p] += sales[b][p][w];
                branchTotal[b] += sales[b][p][w];
                weekTotal[w] += sales[b][p][w];
            }
        }
    }
    printf("PRODUCT-WISE TOTAL SALES\n");
    for (b = 0; b < 2; b++) {
        printf("\nBranch %d:\n", b + 1);

        for (p = 0; p < 3; p++) {
            printf("Product %d: %d\n",
                   p + 1, productTotal[b][p]);
        }
    }
    printf("\nHIGHEST-SELLING PRODUCT IN EACH BRANCH\n");
```


3d.c
3D_2.c
hospital.c
f1.c
f2.c
f3.c
f4.c
f5.c
f6.c
f7.c
folderofc.c > practice.c > 3D_2.c
3 int main() {
31 for (b = 0; b < 2; b++) {
38 }
39 printf("\nHIGHEST-SELLING PRODUCT IN EACH BRANCH\n");
40 for (b = 0; b < 2; b++) {
41 highestProduct = 0;
42 for (p = 1; p < 3; p++) {
43 if (productTotal[b][p] >
44 productTotal[b][highestProduct]) {
45 highestProduct = p;
46 }
47 }
48 printf("Branch %d: Product %d (%d units)\n",
49 b + 1, highestProduct + 1,
50 productTotal[b][highestProduct]);
51 }
52 highestWeek = 0;
53 for (w = 1; w < 4; w++) {
54 if (weekTotal[w] > weekTotal[highestWeek]) {
55 highestWeek = w;
56 }
57 }
58 printf("\nWEEK WITH HIGHEST OVERALL SALES\n");
59 printf("Week %d: %d units\n",highestWeek + 1, weekTotal[highestWeek]);
60 highestBranch = 0;
61 for (b = 1; b < 2; b++) {
62 if (branchTotal[b] > branchTotal[highestBranch]) {
63 highestBranch = b;
64 }
65 }
66 printf("\nTOTAL SALES OF EACH BRANCH\n");
67 for (b = 0; b < 2; b++) {
68 printf("Branch %d: %d units\n",
69 b + 1, branchTotal[b]);
70 }
71 printf("\nBRANCH WITH GREATEST SALES\n");
72 printf("Branch %d: %d units\n",highestBranch + 1, branchTotal[highestBranch]);
73 return 0;
74 }

```
Branch 1:
Product 1: 70
Product 2: 80
Product 3: 58

Branch 2:
Product 1: 90
Product 2: 100
Product 3: 70

HIGHEST-SELLING PRODUCT IN EACH BRANCH
Branch 1: Product 2 (80 units)
Branch 2: Product 2 (100 units)

WEEK WITH HIGHEST OVERALL SALES
Week 4: 159 units

TOTAL SALES OF EACH BRANCH
Branch 1: 208 units
Branch 2: 260 units

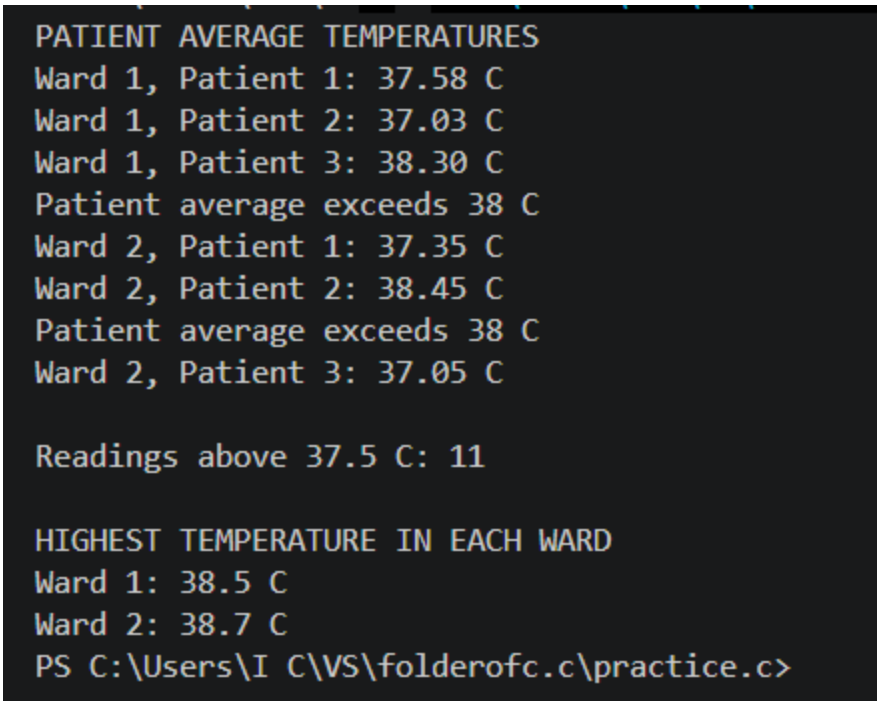
BRANCH WITH GREATEST SALES
Branch 2: 260 units
PS C:\Users\I C\VS\folderofc.c\practice.c>
```

3. A hospital monitors 2 wards, each containing 3 patients. The temperature of each patient is recorded 4 times. Store the readings in a 3D array and write a C program to calculate each patient's average temperature, count readings above 37.5°C, find the highest temperature recorded in each ward, and identify patients whose average temperature exceeds 38°C

folderofc.c > practice.c > C hospital.c

```c
int main() {
    float temp[2][3][4] = {
        {
            {37.0, 37.5, 38.0, 37.8},
            {36.8, 37.0, 37.2, 37.1},
            {38.2, 38.5, 38.1, 38.4}
        },
        {
            {37.1, 37.4, 37.6, 37.3},
            {38.5, 38.2, 38.7, 38.4},
            {36.9, 37.0, 37.2, 37.1}
        }
    };
    int w, p, r, count = 0;
    float sum, avg, highest;
    printf("PATIENT AVERAGE TEMPERATURES\n");
    for (w = 0; w < 2; w++) {
        for (p = 0; p < 3; p++) {
            sum = 0;
            for (r = 0; r < 4; r++) {
                sum += temp[w][p][r];

                if (temp[w][p][r] > 37.5) {
                    count++;
                }
            }
            avg = sum / 4.0;
            printf("Ward %d, Patient %d: %.2f C\n",w + 1, p + 1, avg);
            if (avg > 38.0) {
                printf("Patient average exceeds 38 C\n");
            }
        }
    }
    printf("\nReadings above 37.5 C: %d\n", count);
    printf("\nHIGHEST TEMPERATURE IN EACH WARD\n");
    for (w = 0; w < 2; w++) {
        highest = temp[w][0][0];
        for (p = 0; p < 3; p++) {
            for (r = 0; r < 4; r++) {
                if (temp[w][p][r] > highest) {
                    highest = temp[w][p][r];
                }
            }
        }
        printf("Ward %d: %.1f C\n", w + 1, highest);
    }
```



4. Write a C program that calculates an electricity bill based on units consumed and customer category. Domestic customers are charged Rs. 15 per unit for the first 100 units and Rs. 20 per unit for additional units, while commercial customers are charged Rs. 25 per unit for all units consumed. If the bill exceeds Rs. 5,000, apply a 5% surcharge. Display the units consumed, bill amount, surcharge, and final payable amount.

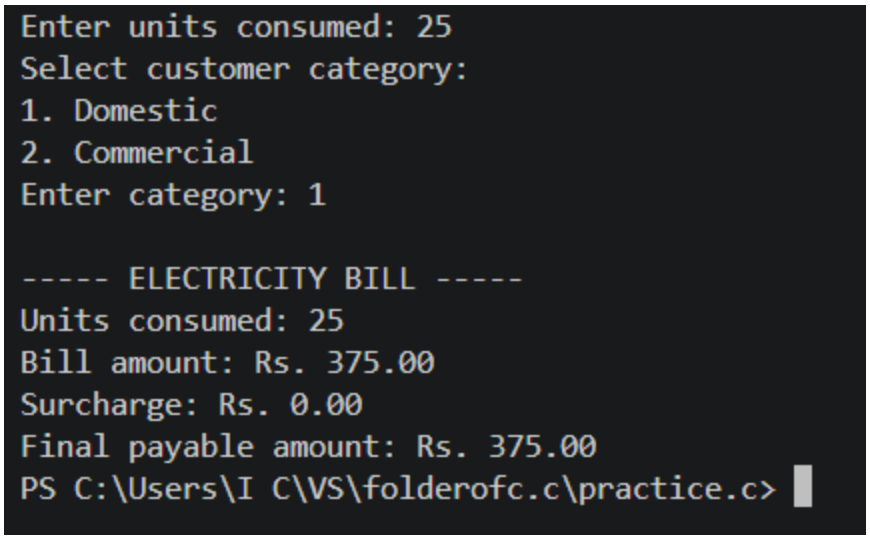
3d.c | 3D_2.c | hospital.c | f1.c | f2.c | f3.c | f4.c | f5.c | f6.c

folderofc.c > practice.c > f1.c

```c
#include <stdio.h>
float calculateBill(int units, int category) {
    float bill;
    if (category == 1) {
        if (units <= 100)
            bill = units * 15;
        else
            bill = (100 * 15) + ((units - 100) * 20);
    }
    else {
        bill = units * 25;
    }

    return bill;
}
int main() {
    int units, category;
    float bill, surcharge = 0, finalBill;
    printf("Enter units consumed: ");
    scanf("%d", &units);
    printf("Select customer category:\n");
    printf("1. Domestic\n");
    printf("2. Commercial\n");
    printf("Enter category: ");
    scanf("%d", &category);
    if (units < 0 || (category != 1 && category != 2)) {
        printf("Invalid input!\n");
        return 1;
    }
    bill = calculateBill(units, category);
    if (bill > 5000){
        surcharge = bill * 0.05;
    }
    finalBill = bill + surcharge;
    printf("\n----- ELECTRICITY BILL -----\n");
    printf("Units consumed: %d\n", units);
    printf("Bill amount: Rs. %.2f\n", bill);
    printf("Surcharge: Rs. %.2f\n", surcharge);
    printf("Final payable amount: Rs. %.2f\n", finalBill);

    return 0;
}
```



5. Write a C program that calculates an employee's final salary based on basic salary and years of service. Employees with more than 5 years of service receive a 15% bonus, those with 3–5 years receive a 10% bonus, and those with less than 3 years receive a 5% bonus. If the salary after adding the bonus exceeds Rs. 100,000, deduct 5% tax. Display the basic salary, bonus, tax, and final salary.

folderofc.c > practice.c > C f2.c

```c
#include <stdio.h>
float calculateBonus(float salary, int years) {
    float bonus;
    if (years > 5)
        bonus = salary * 0.15;
    else if (years >= 3)
        bonus = salary * 0.10;
    else
        bonus = salary * 0.05;

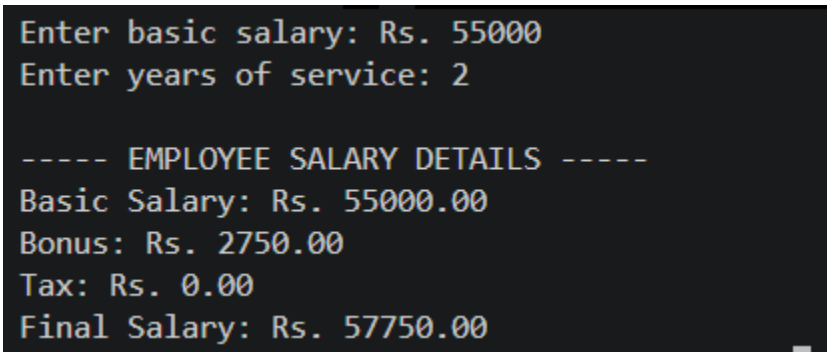
    return bonus;
}
int main() {
    float basicSalary, bonus, salaryAfterBonus;
    float tax = 0, finalSalary;
    int years;

    printf("Enter basic salary: Rs. ");
    scanf("%f", &basicSalary);

    printf("Enter years of service: ");
    scanf("%d", &years);

    if (basicSalary < 0 || years < 0) {
        printf("Invalid input!\n");
        return 1;
    }
    bonus = calculateBonus(basicSalary, years);
    salaryAfterBonus = basicSalary + bonus;
    if(salaryAfterBonus > 100000)
        tax = salaryAfterBonus * 0.05;
    finalSalary = salaryAfterBonus - tax;
    printf("\n----- EMPLOYEE SALARY DETAILS -----\n");
    printf("Basic Salary: Rs. %.2f\n", basicSalary);
    printf("Bonus: Rs. %.2f\n", bonus);
    printf("Tax: Rs. %.2f\n", tax);
    printf("Final Salary: Rs. %.2f\n", finalSalary);

    return 0;
```



6. Write a C program that determines whether a person qualifies for a loan based on age, monthly salary, and existing monthly loan payments. The applicant must be at least 21 years old, earn a monthly salary of at least Rs. 50,000, and have at least 40% of their salary remaining after existing payments. If eligible, calculate the maximum loan amount as 10 times the remaining monthly income. Otherwise, display the reason for ineligibility

folderofc.c > practice.c > C f3.c

```c
#include <stdio.h>
float calculateRemaining(float salary, float payments) {
    return salary - payments;
}
int main() {
    int age;
    float salary, payments, remaining, loanAmount;
    printf("Enter your age: ");
    scanf("%d", &age);
    printf("Enter monthly salary: Rs. ");
    scanf("%f", &salary);
    printf("Enter existing monthly loan payments: Rs. ");
    scanf("%f", &payments);
    if (age < 0 || salary < 0 || payments < 0){
        printf("Invalid input!\n");
        return 1;
    }
    remaining = calculateRemaining(salary, payments);
    printf("\n----- LOAN ELIGIBILITY -----\n");
    if (age < 21){
        printf("Not eligible: Age must be at least 21.\n");
    }
    else if (salary < 50000){
        printf("Not eligible: Monthly salary must be at least Rs. 50,000.\n");
    }
    else if (remaining < salary * 0.40) {
        printf("Not eligible: At least 40%% of salary must remain after payments.\n");
    }
    else{
        loanAmount = remaining * 10;
        printf("Status: Eligible for loan\n");
        printf("Monthly salary: Rs. %.2f\n", salary);
        printf("Remaining monthly income: Rs. %.2f\n", remaining);
        printf("Maximum loan amount: Rs. %.2f\n", loanAmount);
    }
    return 0;
}
```

```
Enter your age: 25
Enter monthly salary: Rs. 25000
Enter existing monthly loan payments: Rs. 2500

----- LOAN ELIGIBILITY -----
Not eligible: Monthly salary must be at least Rs. 50,000.
PS C:\Users\I C\VS\folderofc.c\practice.c>
```

7. Write a C program that accepts a positive integer and determines whether it is even or odd, whether it is a palindrome, and whether it is a prime number. A palindrome number remains the same when its digits are reversed. Use arithmetic operations and loops to perform the checks without converting the number into a string or using arrays. Display all results.

folderofc.c > practice.c > C f4.c

```c
#include <stdio.h>
void checkEvenOdd(int n) {
    if (n % 2 == 0)
        printf("Number is Even\n");
    else
        printf("Number is Odd\n");
}
void checkPalindrome(int n) {
    int original = n;
    int reverse = 0, remainder;
    while (n > 0) {
        remainder = n % 10;
        reverse = reverse * 10 + remainder;
        n = n / 10;
    }
    if (original == reverse)
        printf("Number is Palindrome\n");
    else
        printf("Number is Not Palindrome\n");
}
void checkPrime(int n) {
    int i, isPrime = 1;
    if (n == 1)
        isPrime = 0;
    for (i = 2; i <= n / i; i++) {
        if (n % i == 0) {
            isPrime = 0;
            break;
        }
    }
    if (isPrime == 1)
```

```
    }
    if (isPrime == 1)
        printf("Number is Prime\n");
    else
        printf("Number is Not Prime\n");
}

int main() {
    int n;

    printf("Enter a positive integer: ");
    scanf("%d", &n);

    if (n <= 0) {
        printf("Please enter a positive integer.\n");
        return 1;
    }

    checkEvenOdd(n);
    checkPalindrome(n);
    checkPrime(n);

    return 0;
}
```

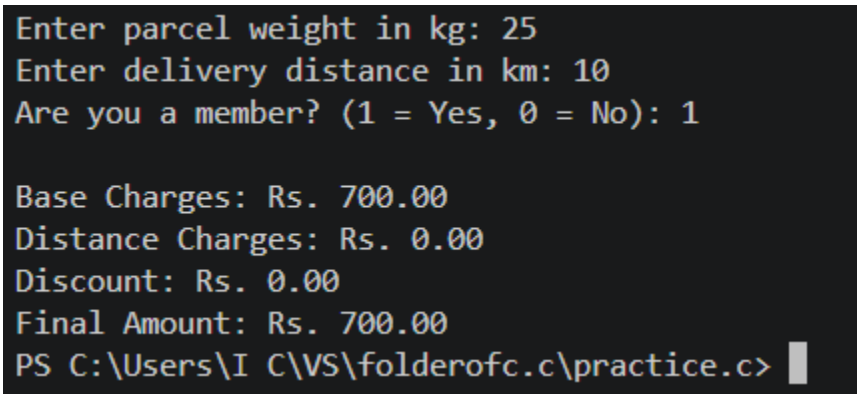
```
Enter a positive integer: 25
Number is Odd
Number is Not Palindrome
Number is Not Prime
PS C:\Users\I C\VS\folderofc.c\practice.c>
```

8. Write a C program that calculates delivery charges based on parcel weight, delivery distance, and customer membership. Parcels weighing up to 2 kg cost Rs. 200, those weighing above 2 kg and up to 5 kg cost Rs. 400, and those weighing above 5 kg cost Rs. 700. Add Rs. 20 per kilometre for distances exceeding 10 km. Members receive a 10% discount if the total charges exceed Rs. 1,000. Display the base charges, distance charges, discount, and final amount.

folderofc.c > practice.c > C f5.c

```c
#include <stdio.h>
float getBaseCharges(float weight) {
    if (weight <= 2)
        return 200;
    else if (weight <= 5)
        return 400;
    else
        return 700;
}
float getDistanceCharges(float distance) {
    if (distance > 10)
        return (distance - 10) * 20;
    else
        return 0;
}
float getDiscount(float total, int member) {
    if (member == 1 && total > 1000)
        return total * 0.10;
    else
        return 0;
}
int main() {
    float weight, distance;
    float base, distanceCharges, total, discount, finalAmount;
    int member;
    printf("Enter parcel weight in kg: ");
    scanf("%f", &weight);
    printf("Enter delivery distance in km: ");
    scanf("%f", &distance);
    printf("Are you a member? (1 = Yes, 0 = No): ");
    scanf("%d", &member);
    base = getBaseCharges(weight);
    distanceCharges = getDistanceCharges(distance);
    total = base + distanceCharges;
    discount = getDiscount(total, member);
    finalAmount = total - discount;
    printf("\nBase Charges: Rs. %.2f", base);
    printf("\nDistance Charges: Rs. %.2f", distanceCharges);
    printf("\nDiscount: Rs. %.2f", discount);
    printf("\nFinal Amount: Rs. %.2f\n", finalAmount);

    return 0;
}
```



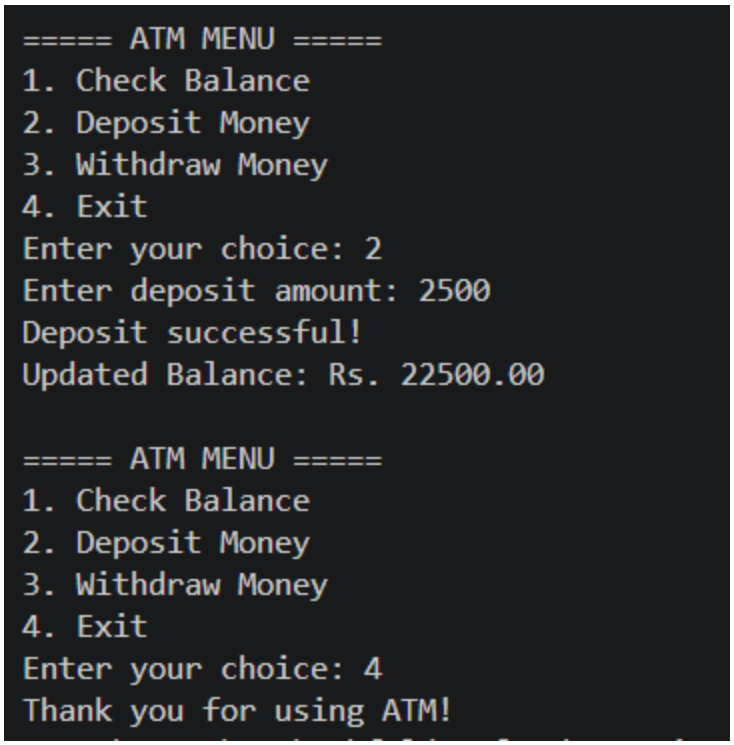
9. Write a menu-driven C program that starts with a balance of Rs. 20,000 and allows the user to check the balance, deposit money, withdraw money, or exit. Deposits and withdrawals must be positive, and withdrawals cannot exceed the available balance or Rs. 10,000 per transaction. If the balance falls below Rs. 5,000 after a withdrawal, display a low-balance warning. Continue displaying the menu until the user selects Exit, and display the updated balance after every successful transaction

```c
#include <stdio.h>
void checkBalance(float balance) {
    printf("Current Balance: Rs. %.2f\n", balance);
}
float deposit(float balance) {
    float amount;
    printf("Enter deposit amount: ");
    scanf("%f", &amount);
    if (amount > 0) {
        balance = balance + amount;
        printf("Deposit successful!\n");
        printf("Updated Balance: Rs. %.2f\n", balance);
    }
    else{
        printf("Amount must be positive!\n");
    }
    return balance;
}
float withdraw(float balance) {
    float amount;
    printf("Enter withdrawal amount: ");
    scanf("%f", &amount);
    if (amount <= 0) {
        printf("Amount must be positive!\n");
    } else if (amount > 10000) {
        printf("Maximum withdrawal limit is Rs. 10,000.\n");
    } else if (amount > balance) {
        printf("Insufficient balance!\n");
    } else {
        balance = balance - amount;
        printf("Withdrawal successful!\n");
        printf("Updated Balance: Rs. %.2f\n", balance);

        if (balance < 5000){
            printf("Warning: Low balance!\n");
```

```c
            printf("Warning: Low balance!\n");
        }
    }
    return balance;
}
int main() {
    float balance = 20000;
    int choice;
    do {
        printf("\n===== ATM MENU =====\n");
        printf("1. Check Balance\n");
        printf("2. Deposit Money\n");
        printf("3. Withdraw Money\n");
        printf("4. Exit\n");
        printf("Enter your choice: ");
        scanf("%d", &choice);
        switch (choice) {
            case 1:
                checkBalance(balance);
                break;
            case 2:
                balance = deposit(balance);
                break;
            case 3:
                balance = withdraw(balance);
                break;
            case 4:
                printf("Thank you for using ATM!\n");
                break;
            default:
                printf("Invalid choice!\n");
        }

    } while (choice != 4);
    return 0;
```



10. Write a C program that calculates a shopping bill using product price, quantity, and customer membership status. If the subtotal exceeds Rs. 5,000, apply a 10% discount. Members receive an additional 5% discount on the subtotal. After applying the discounts, add a 2% payment fee if the customer pays by card. Display the subtotal, each applicable discount, payment fee, and final payable amount.

folderofc.c > practice.c > C f7.c

```c
#include <stdio.h>
float getDiscount(float subtotal) {
    if (subtotal > 5000)
        return subtotal * 0.10;
    else
        return 0;
}
float getMemberDiscount(float subtotal, int member) {
    if (member == 1)
        return subtotal * 0.05;
    else
        return 0;
}
float getPaymentFee(float amount, int payment) {
    if (payment == 1)
        return amount * 0.02;
    else
        return 0;
}
int main() {
    float price, subtotal, discount, memberDiscount;
    float paymentFee, finalAmount;
    int quantity, member, payment;
    printf("Enter product price: ");
    scanf("%f", &price);
    printf("Enter quantity: ");
    scanf("%d", &quantity);
    printf("Are you a member? (1 = Yes, 0 = No): ");
    scanf("%d", &member);
    printf("Payment method (1 = Card, 0 = Cash): ");
    scanf("%d", &payment);
    subtotal = price * quantity;
    discount = getDiscount(subtotal);
    memberDiscount = getMemberDiscount(subtotal, member);
    finalAmount = subtotal - discount - memberDiscount;
    paymentFee = getPaymentFee(finalAmount, payment);
    finalAmount = finalAmount + paymentFee;
    printf("\nSubtotal: Rs. %.2f\n", subtotal);
    printf("Discount: Rs. %.2f\n", discount);
    printf("Member Discount: Rs. %.2f\n", memberDiscount);
    printf("Payment Fee: Rs. %.2f\n", paymentFee);
    printf("Final Payable Amount: Rs. %.2f\n", finalAmount);

    return 0;
}
```

```
Enter product price: 2500
Enter quantity: 4
Are you a member? (1 = Yes, 0 = No): 1
Payment method (1 = Card, 0 = Cash): 0

Subtotal: Rs. 10000.00
Discount: Rs. 1000.00
Member Discount: Rs. 500.00
Payment Fee: Rs. 0.00
Final Payable Amount: Rs. 8500.00
PS C:\Users\I C\VS\folderofc.c\practice.c>
```